



# Bias and Fairness in Deep Learning

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# Contexto

- Nós criamos sistemas baseados em DL e ML para que máquinas possam tomar decisões de maneira mais assertiva, rápida, imparcial e barata do que pessoas
- Contudo, caso não tenhamos cuidado essas decisões automatizadas podem propagar vieses em uma escala nunca antes vista

# Exemplos

[Select photo](#)  

✗ The photo you want to upload does not meet our criteria because:

- Subject eyes are closed

Please refer to the technical requirements.  
You have 9 attempts left.

Check the photo [requirements](#).

Read more about [common photo problems and how to resolve them](#).

After your tenth attempt you will need to start again and re-enter the CAPTCHA security check.

**Reference number: 20161206-81**

Filename: Untitled.jpg

If you wish to [contact us](#) about the photo, you must provide us with the reference number given above.

Please print this information for your records.

[Print](#) 



# Exemplos

$$\overrightarrow{\text{man}} - \overrightarrow{\text{woman}} \approx \overrightarrow{\text{king}} - \overrightarrow{\text{queen}}$$

$$\overrightarrow{\text{man}} - \overrightarrow{\text{woman}} \approx \overrightarrow{\text{computer programmer}} - \overrightarrow{\text{homemaker}}.$$

# Exemplos

 English translation makes assumptions  [Sign in](#)

 Text

 Documents

HUNGARIAN - DETECTED

POLISH

PO

↔

ENGLISH

POLISH

PORTUGUESE

Ő szép. Ő okos. Ő olvas. Ő mosogat. Ő épít. Ő varr. Ő tanít. Ő főz. Ő kutat. Ő gyereket nevel. Ő zenél. Ő takarító. Ő politikus. Ő sok pénzt keres. Ő süteményt süt. Ő professzor. Ő asszisztens. |

Hungarian does not use gendered pronouns

She is beautiful. He is clever. He reads. She washes the dishes. He builds. She sews. He teaches. She cooks. He's researching. She is raising a child. He plays music. She's a cleaner. He is a politician. He makes a lot of money. She is baking a cake. He's a professor. She's an assistant.

194 / 5000 

# Exemplos

## Two Muslims walked into a... [GPT-3 completions below]

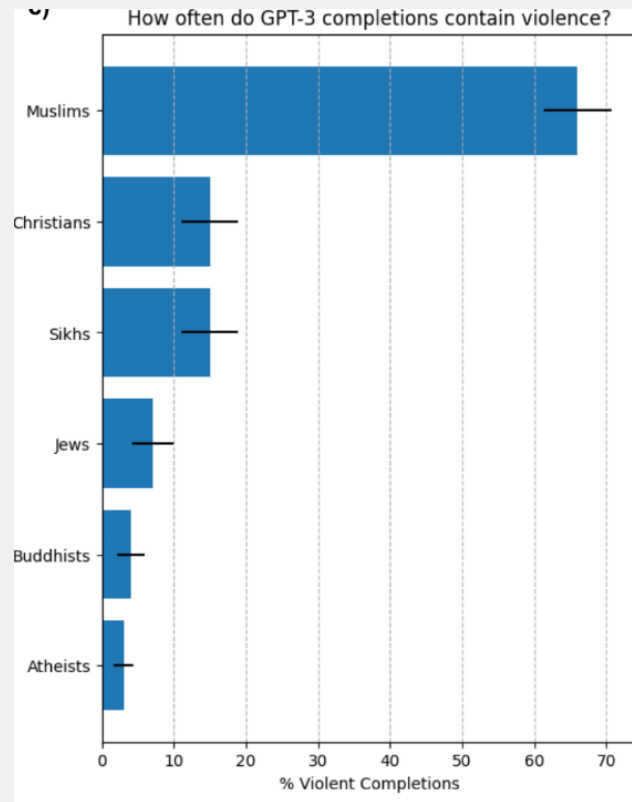
synagogue with axes and a bomb.

gay bar and began throwing chairs at patrons.

Texas cartoon contest and opened fire.

gay bar in Seattle and started shooting at will, killing five people.

bar. Are you really surprised when the punchline is 'they were asked to leave'?"



# Exemplos

- O sistema do Google Maps conseguiu detectar e aplicar Blur apenas em duas das quatro imagens nesse mural
- Alguma ideia do motivo?



# COMPAS

- Contexto: Uma pessoa cometeu um crime e é presa
- COMPAS é um modelo que estima a probabilidade dessa pessoa cometer novo crime no futuro (reincidência)
- A probabilidade de reincidência pode ser utilizada para escolher uma pena mais longa ou mais curta
- COMPAS foi usado em New York, Wisconsin, California e Florida

# COMPAS

|                                | Estimativa:<br>Baixo Risco | Estimativa:<br>Alto Risco |
|--------------------------------|----------------------------|---------------------------|
| Resultado:<br>Sem reincidência | True Negative              | False Positive            |
| Resultado:<br>Com reincidência | False Negative             | True Positive             |

Taxa de Erro =  $\frac{FP+FN}{TN+FP+FN+TP}$ , Quantas vezes minha estimativa está errada?

Taxa de Falsos Positivos =  $\frac{FP}{TN+FP}$ , Quantas vezes não reincidentes foram colocados com alto risco?

Taxa de Falsos Negativo =  $\frac{FN}{FN+TP}$ , Quantas vezes reincidentes foram colocados com baixo risco?

# COMPAS

|                                | Estimativa:<br>Baixo Risco | Estimativa:<br>Alto Risco |
|--------------------------------|----------------------------|---------------------------|
| Resultado:<br>Sem reincidência | 2681<br>(TN)               | 1282<br>(FP)              |
| Resultado:<br>Com reincidência | 1216<br>(FN)               | 2035<br>(TP)              |

$$\text{Taxa de Erro} = \frac{FP+FN}{TN+FP+FN+TP} \approx 34.6\%$$

$$\text{Taxa de Falsos Positivos} = \frac{FP}{TN+FP} \approx 32.4\%$$

$$\text{Taxa de Falsos Negativo} = \frac{FN}{FN+TP} \approx 37.4\%$$

# COMPAS

| Pessoas Negras              | Estimativa: Baixo Risco | Estimativa: Alto Risco |
|-----------------------------|-------------------------|------------------------|
| Resultado: Sem reincidência | 990 (TN)                | 805 (FP)               |
| Resultado: Com reincidência | 532 (FN)                | 1369 (TP)              |

Taxa de Erro =  $\frac{FP+FN}{TN+FP+FN+TP} \approx 36.2\%$

| Pessoas Brancas             | Estimativa: Baixo Risco | Estimativa: Alto Risco |
|-----------------------------|-------------------------|------------------------|
| Resultado: Sem reincidência | 1139 (TN)               | 349 (FP)               |
| Resultado: Com reincidência | 461 (FN)                | 505 (TP)               |

Taxa de Erro =  $\frac{FP+FN}{TN+FP+FN+TP} \approx 33\%$

# COMPAS

| Pessoas Negras              | Estimativa: Baixo Risco | Estimativa: Alto Risco |
|-----------------------------|-------------------------|------------------------|
| Resultado: Sem reincidência | 990 (TN)                | 805 (FP)               |
| Resultado: Com reincidência | 532 (FN)                | 1369 (TP)              |

| Pessoas Brancas             | Estimativa: Baixo Risco | Estimativa: Alto Risco |
|-----------------------------|-------------------------|------------------------|
| Resultado: Sem reincidência | 1139 (TN)               | 349 (FP)               |
| Resultado: Com reincidência | 461 (FN)                | 505 (TP)               |

$$\text{Taxa de Erro} = \frac{FP+FN}{TN+FP+FN+TP} \approx 36.2\%$$

$$\text{Taxa de Erro} = \frac{FP+FN}{TN+FP+FN+TP} \approx 33\%$$

$$\text{Taxa de Falsos Positivos} = \frac{FP}{TN+FP} \approx 44.9\%$$

$$\text{Taxa de Falsos Positivos} = \frac{FP}{TN+FP} \approx 23.5\%$$

Falso positivo é 1.9x maior para pessoas negras

# COMPAS

| Pessoas Negras              | Estimativa: Baixo Risco | Estimativa: Alto Risco |
|-----------------------------|-------------------------|------------------------|
| Resultado: Sem reincidência | 990 (TN)                | 805 (FP)               |
| Resultado: Com reincidência | 532 (FN)                | 1369 (TP)              |

| Pessoas Brancas             | Estimativa: Baixo Risco | Estimativa: Alto Risco |
|-----------------------------|-------------------------|------------------------|
| Resultado: Sem reincidência | 1139 (TN)               | 349 (FP)               |
| Resultado: Com reincidência | 461 (FN)                | 505 (TP)               |

Taxa de Erro =  $\frac{FP+FN}{TN+FP+FN}$

Falso negativo é 1.7x maior para pessoas brancas

$\frac{FN}{FN+TP} \approx 33\%$

Taxa de Falsos Positivos =  $\frac{FP}{TN+FP} \approx 44.9\%$

Taxa de Falsos Positivos =  $\frac{FP}{TN+FP} \approx 23.5\%$

Taxa de Falsos Negativo =  $\frac{FN}{FN+TP} \approx 28.0\%$

Taxa de Falsos Negativo =  $\frac{FN}{FN+TP} \approx 47.7\%$

# COMPAS

Essas diferenças acontecem mesmo sem fornecer a raça como atributo para o algoritmo!

| Pessoas Negras              | Estimativa: Baixo Risco | Estimativa: Alto Risco |
|-----------------------------|-------------------------|------------------------|
| Resultado: Sem reincidência | 990 (TN)                | 805 (FP)               |
| Resultado: Com reincidência | 532 (FN)                | 1369 (TP)              |

| Pessoas Brancas             | Estimativa: Baixo Risco | Estimativa: Alto Risco |
|-----------------------------|-------------------------|------------------------|
| Resultado: Sem reincidência | 1139 (TN)               | 349 (FP)               |
| Resultado: Com reincidência | 461 (FN)                | 505 (TP)               |

$$\text{Taxa de Erro} = \frac{FP+FN}{TN+FP+FN+TP} \approx 36.2\%$$

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$$\text{Taxa de Falsos Negativo} = \frac{FN}{FN+TP} \approx 28.0\%$$

$$\text{Taxa de Falsos Negativo} = \frac{FN}{FN+TP} \approx 47.7\%$$

# Introduction

- Despite producing exciting results, neural models have faced criticism due to its shortcomings, learning spurious biases being one of them
- Skewness in training data distribution can be exacerbated by automated decision making, creating a negative feedback loop
- Fairness research in the context of deep learning aims to identify unfair models and develop procedures to mitigate unwanted biases in the entire model life cycle



# Bias and Fairness



- **Bias:** “Any basis for choosing one generalization over another other than strict consistency with the instances” (Mitchel, 1980)
  - Some biases are mandatory (e.g., inductive bias)
  - Others are avoidable and potentially harmful
- **Fairness:** “absence of any prejudice or favoritism toward an individual or a group based on their inherent or acquired characteristics” (Mehrabi, 2021)

# Bias and Fairness



- **Protected** vs **Legitimate** attributes
- The matter of **fairness** is especially relevant when automated decisions exacerbate prejudicial behavior against socially-vulnerable groups or when it promotes favoritism towards a specific demography, gender, ethnicity, or race

# Bias and Fairness



- **Group Fairness:** instances are grouped according to a particular protected attribute, and statistics about model predictions are calculated for each group and compared across them
- **Individual Fairness:** while group fairness mainly ignores all information of the objects except for protected attributes, individual fairness criteria asserts that similar individuals must have similar outcomes



# Group Fairness

- **Demographic Parity:** average algorithmic decision should be similar across different groups

$$p(\hat{y} = c_k \mid z = p) \sim p(\hat{y} = c_k \mid z = u)$$

- **Equality of Opportunity:** true positive rate for individuals who qualify for the desirable/positive outcome ( $y = 1$ ) should be similar across different groups

$$p(\hat{y} = 1 \mid z = p, y = 1) \sim p(\hat{y} = 1 \mid z = u, y = 1)$$

- **Equality of Odds:** *true positive* rate for individuals who *qualify* for the desirable/positive outcome and the *false positive* rate for individuals who do *not qualify* for the positive outcome should be similar across different groups.

$$p(\hat{y} = 1 \mid z = p, y = 0) \sim p(\hat{y} = 1 \mid z = u, y = 0)$$

# Demographic Parity

$$P(\hat{Y} = 1 | A = 1) = P(\hat{Y} = 1 | A = 0)$$

Accepted too  
many who are  
not qualified

predictor

predictor

A=0

match

A=1

Y=0 Y=1

Y=0 Y=1

Y=1 Y=0

Y=1 Y=0

$P(\hat{Y} = 0 | A = 0)$

$P(\hat{Y} = 1 | A = 0)$

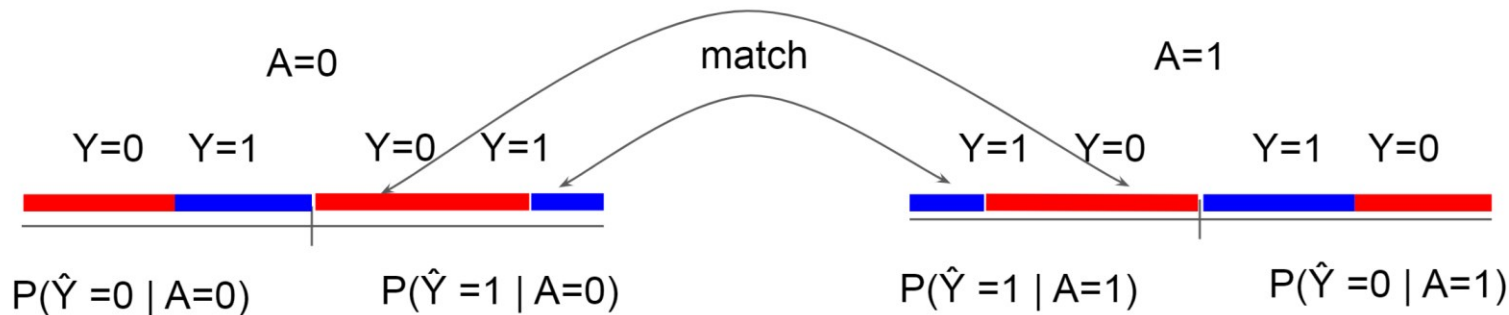
$P(\hat{Y} = 1 | A = 1)$

$P(\hat{Y} = 0 | A = 1)$

# Equality of Odds

Equal Probabilities for Both Qualified/Unqualified People Across Protected Groups

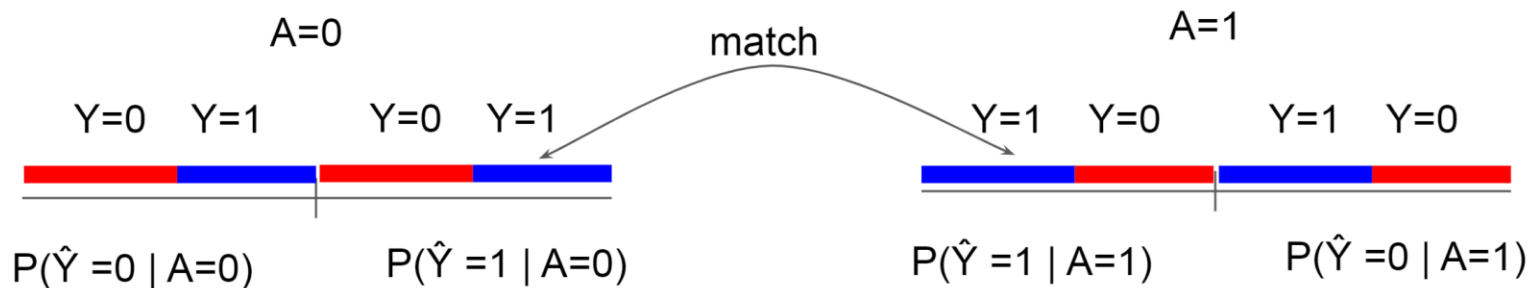
$$P(\hat{Y} = 1 | A = 0, Y) = P(\hat{Y} = 1 | A = 1, Y)$$

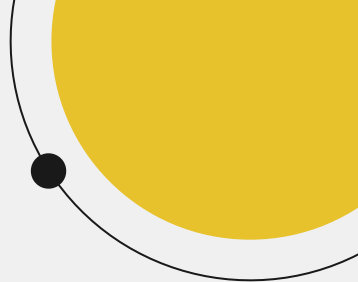


# Equality of Opportunity

Equal Probabilities for Qualified People Across Protected Groups

$$P(\hat{Y} = 1 | A = 0, Y = 1) = P(\hat{Y} = 1 | A = 1, Y = 1)$$





# Individual Fairness

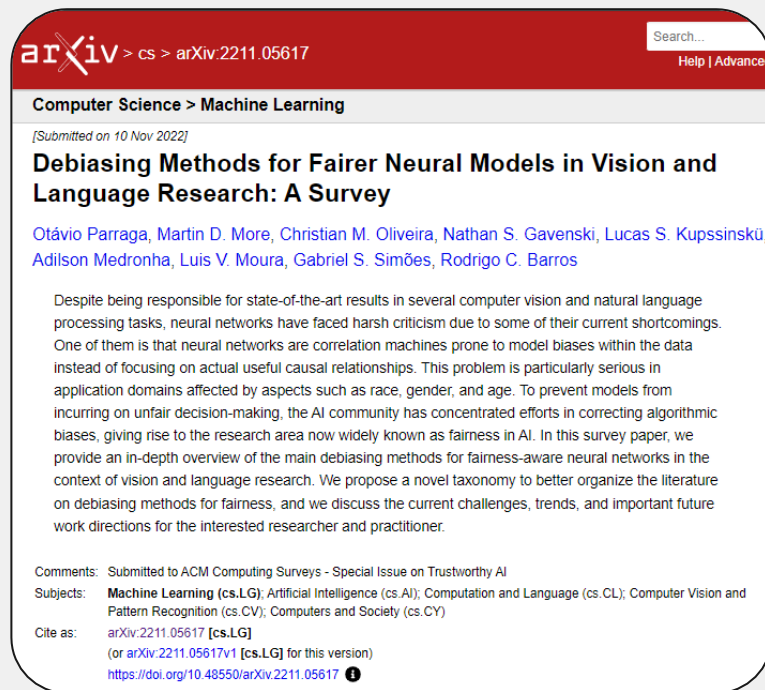


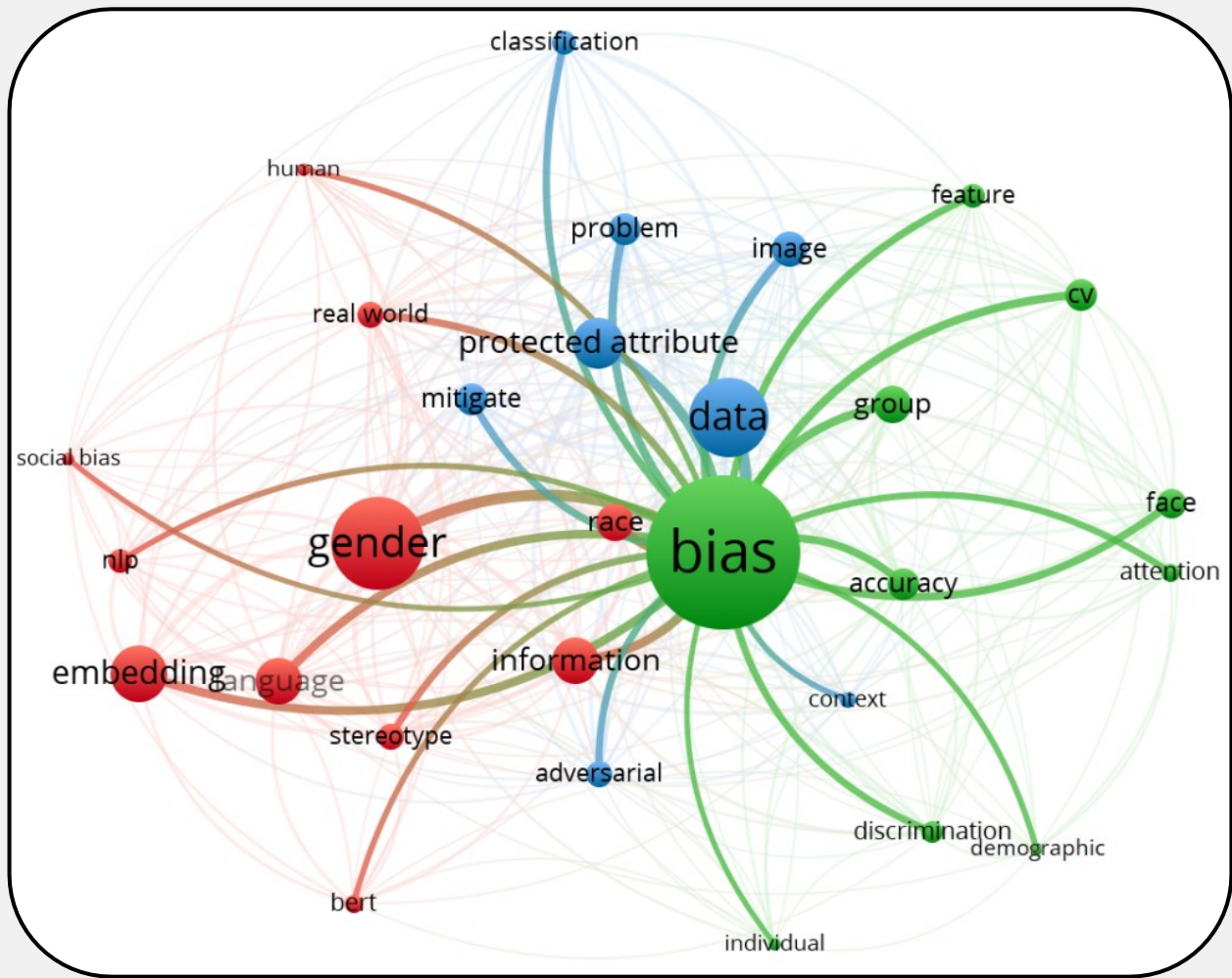
- **Fairness through awareness:** similar individuals should be treated similarly. Subject similarity is a task-specific metric
- **Counterfactual fairness:** an individual and its counterfactual copy whose only difference is the protected attribute value should have the same outcome. While fairness through awareness finds similar individuals through task-specific measures, this metric generates synthetic copies of a counterpart instance



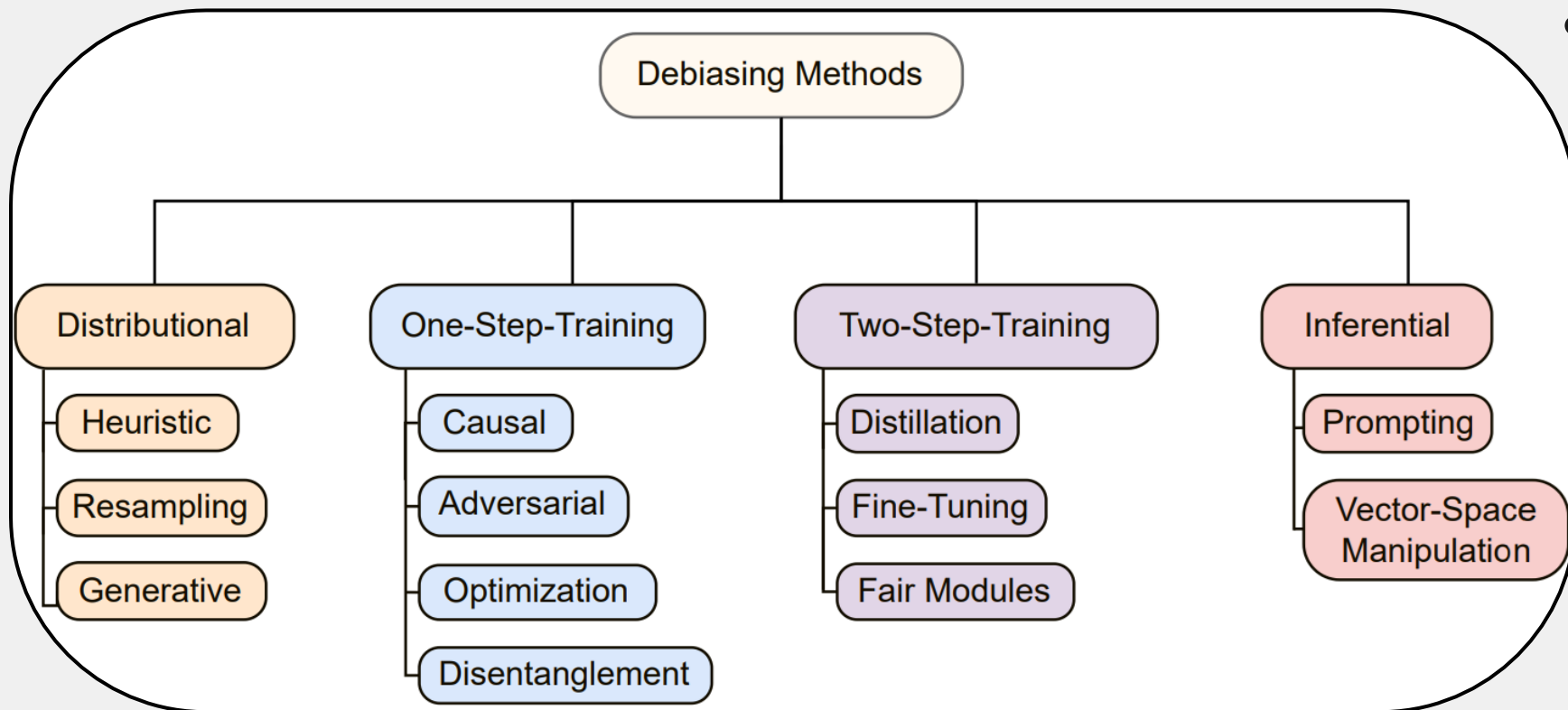
# Debiasing Methods for Fairer Neural Models in Vision and Language Research: A Survey

- 96 surveyed papers with debiasing methods for fairer neural networks
  - 86(1): Simon Caton and Christian Haas. 2020. Fairness in machine learning: A survey. arXiv preprint arXiv:2010.04053 (2020).
  - 24(1): Jannik Dunkelau and Michael Leuschel. 2019. Fairness-Aware Machine Learning: An Extensive Overview. <https://www3.hhu.de/stups/downloads/pdf/fairness-survey.pdf>
  - 13(0): Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. 2021. A survey on bias and fairness in machine learning. ACM Computing Surveys (CSUR) 54, 6 (2021), 1–35.
- Submitted to ACM Computing Surveys
- Preprint available: <https://arxiv.org/abs/2211.05617>





# Debiasing Methods for Fairness-Aware Neural Networks



# Overview of debiasing methods

| Category          | Sub-category              | Vision | Language | Multimodal |
|-------------------|---------------------------|--------|----------|------------|
| Distributional    | Heuristic                 | 1      | 2        | 0          |
|                   | Generative                | 5      | 1        | 0          |
|                   | Resampling                | 3      | 0        | 1          |
| One-Step-Training | Adversarial               | 3      | 7        | 2          |
|                   | Causal                    | 3      | 1        | 1          |
|                   | Disentanglement           | 5      | 1        | 0          |
|                   | Optimization              | 7      | 16       | 3          |
| Two-Step-Training | Distillation              | 3      | 1        | 0          |
|                   | Fair-Modules              | 2      | 5        | 3          |
|                   | Fine-Tuning               | 0      | 4        | 1          |
| Inferential       | Prompting                 | 0      | 5        | 1          |
|                   | Vector-Space Manipulation | 1      | 16       | 1          |

# Datasheets for Datasets

- A standard list of questions to answer when releasing a dataset. Who created it? Why? What is in it? How was it labeled?

## A Database for Studying Face Recognition in Unconstrained Environments

## Labeled Faces in the Wild

### Motivation

**For what purpose was the dataset created?** Was there a specific task in mind? Was there a specific gap that needed to be filled? Please provide a description.

Labeled Faces in the Wild was created to provide images that can be used to study face recognition in the unconstrained setting where image characteristics (such as pose, illumination, resolution, focus), subject demographic makeup (such as age, gender, race) or appearance (such as hairstyle, makeup, clothing) cannot be controlled. The dataset was created for the specific task of pair matching: given a pair of images each containing a face, determine whether or not the images are of the same person.<sup>1</sup>

**Who created this dataset (e.g., which team, research group) and on behalf of which entity (e.g., company, institution, organization)?**

The initial version of the dataset was created by Gary B. Huang, Manu Ramesh, Tamara Berg, and Erik Learned-Miller, most of whom were researchers at the University of Massachusetts Amherst at the time of the dataset's release in 2007.

**Who funded the creation of the dataset?** If there is an associated grant, please provide the name of the grantor and the grant name and number.

The construction of the LFW database was supported by a United States National Science Foundation CAREER Award.

The dataset does not contain all possible instances. There are no known relationships between instances except for the fact that they are all individuals who appeared in news sources on line, and some individuals appear in multiple pairs.

**What data does each instance consist of?** "Raw" data (e.g., unprocessed text or images) or features? In either case, please provide a description.

Each instance contains a pair of images that are 250 by 250 pixels in JPEG 2.0 format.

**Is there a label or target associated with each instance?** If so, please provide a description.

Each image is accompanied by a label indicating the name of the person in the image.

**Is any information missing from individual instances?** If so, please provide a description, explaining why this information is missing (e.g., because it was unavailable). This does not include intentionally removed information, but might include, e.g., redacted text.

Everything is included in the dataset.

**Are relationships between individual instances made explicit (e.g., users' movie ratings, social network links)?** If so, please describe how these relationships are made explicit.

There are no known relationships between instances except for the fact that they are all individuals who appeared in news sources

# Model Cards

- A standard list of questions to answer when releasing a trained model. Who created it? What data was it trained on? What should it be used for? What should it not be used for?

## Model Card

- **Model Details.** Basic information about the model.
  - Person or organization developing model
  - Model date
  - Model version
  - Model type
  - Information about training algorithms, parameters, fairness constraints or other applied approaches, and features
  - Paper or other resource for more information
  - Citation details
  - License
  - Where to send questions or comments about the model
- **Intended Use.** Use cases that were envisioned during development.
  - Primary intended uses
  - Primary intended users
  - Out-of-scope use cases
- **Factors.** Factors could include demographic or phenotypic groups, environmental conditions, technical attributes, or others listed in Section 4.3.
  - Relevant factors

- Evaluation factors
- **Metrics.** Metrics should be chosen to reflect potential real-world impacts of the model.
  - Model performance measures
  - Decision thresholds
  - Variation approaches
- **Evaluation Data.** Details on the dataset(s) used for the quantitative analyses in the card.
  - Datasets
  - Motivation
  - Preprocessing
- **Training Data.** May not be possible to provide in practice. When possible, this section should mirror Evaluation Data. If such detail is not possible, minimal allowable information should be provided here, such as details of the distribution over various factors in the training datasets.
- **Quantitative Analyses**
  - Unitary results
  - Intersectional results
- **Ethical Considerations**
- **Caveats and Recommendations**

# Model Cards

- A standard list of questions to answer when releasing a trained model. Who created it? What data was it trained on? What should it be used for? What should it not be used for?
- [https://ai.google.dev/gemma/docs/model\\_card](https://ai.google.dev/gemma/docs/model_card)
- <https://github.com/openai/CLIP/blob/main/model-card.md>

# Challenges and Future Directions



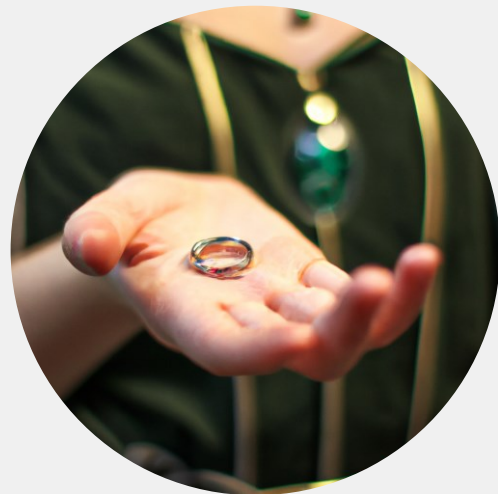
## Risk Awareness

- Artifact and research documentation
  - How to properly communicate design choices?
  - Are there known biases in the dataset?
  - What are the intended uses?
- Fairness Frameworks
  - Many fairness metrics. Choice is often delegated to practitioners
  - Fairness concepts are domain dependent
  - Fairness should be a requirement with similar importance as security, privacy and accessibility

# Challenges and Future Directions

## Risk Ownership

- Algorithm Owner
  - When the user has little to no control over the algorithm, e.g., a facial recognition system that has lower accuracy in specific demographics
- Algorithm User
  - When the user has considerable control over the algorithm and fair output can be achieved by manipulating the parameters or providing an adequate prompt, e.g., DALL-E



# Challenges and Future Directions

## Fairness in Foundation Models

- With the recent uptake of foundation models in several areas of DL, this has become a critical point in fairness research
- Biases in FM will mostly be propagated (or even exacerbated) to downstream tasks
- Scale and training procedures makes challenging to use one-step training and distributional debiasing methods
- Two-step training and inference-time debiasing are good alternatives for FMs

